EASY · ALGEBRA · ANSWER KEY

Linear equations in one variable

01

10 answers · Rationale-rich · Teach from doc

Q1 B

B. 350

Choice B is correct. Subtracting 7 from each side of the given equation yields $w = 350$. Therefore, the value of w that is the solution to the given equation is 350.

Choice A is incorrect. This is the value of w that is the solution to the equation $7w = 357$, not $w + 7 = 357$.

Choice C is incorrect. This is the value of w that is the solution to the equation $w - 7 = 357$, not $w + 7 = 357$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2 A

A. 14

Choice A is correct. Subtracting 180 from both sides of the given equation yields $5p = 70$. Dividing both sides of this equation by 5 yields $p = 14$. Therefore, the value of p that satisfies the equation $5p + 180 = 250$ is 14.

Choice B is incorrect. This value of p satisfies the equation $5p + 180 = 505$.

Choice C is incorrect. This value of p satisfies the equation $5p + 180 = 610$.

Choice D is incorrect. This value of p satisfies the equation $5p + 180 = 1{,}430$.

Q3

C

C. -6

Choice C is correct. Dividing all terms in the given equation by 4 yields $\frac{4x}{4} - \frac{28}{4} = -\frac{24}{4}$, or $x - 7 = -6$. Therefore, the value of $x - 7$ is -6.

Choice A is incorrect. This is the value of $4x - 28$, not $x - 7$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

C

C. 27

Choice C is correct. It’s given that $x = 7$. Substituting 7 for x into the given expression $x + 20$ yields $7 + 20$, which is equivalent to 27.

Choice A is incorrect. This is the value of $x + 6$.

Choice B is incorrect. This is the value of $x + 13$.

Choice D is incorrect. This is the value of $x + 27$.

Q5

C

C. 8

Choice C is correct. It's given that a gym charges its members a onetime $\$36$ enrollment fee and a membership fee of $\$19$ per month. Let x represent the number of months at the gym after which a member will have been charged a total of $\$188$. If there are no charges other than the enrollment fee and the membership fee, the equation $36 + 19x = 188$ can be used to represent this situation. Subtracting 36 from both sides of this equation yields $19x = 152$. Dividing both sides of this equation by 19 yields $x = 8$. Therefore, a member will have been charged a total of $\$188$ after 8 months at the gym.

Choice A is incorrect. A member will have been charged a total of $\$36 + 19 \times 4$, or $\$112$, after 4 months at the gym.

Choice B is incorrect. A member will have been charged a total of $\$36 + 19 \times 5$, or $\$131$, after 5 months at the gym.

Choice D is incorrect. A member will have been charged a total of $\$36 + 19 \times 10$, or $\$226$, after 10 months at the gym.

Q6

A

A. 35

Choice A is correct. It's given that a total of 165 people contributed to a charity event as either a donor or a volunteer. It's also given that 130 people contributed as a donor. It follows that 165 - 130, or 35, people contributed as a volunteer.

Choice B is incorrect. This is the number of people who contributed as a donor, not a volunteer.

Choice C is incorrect. This is the total number of people who contributed as either a donor or a volunteer, not the number of people who contributed as a volunteer.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7

B

B. 25

Choice B is correct. Subtracting 275 from both sides of the given equation yields $2p = 50$. Dividing both sides of this equation by 2 yields $p = 25$. Therefore, the value of p that satisfies the given equation is 25.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of p that satisfies the equation $2 + p + 275 = 325$, not $2p + 275 = 325$.

Choice D is incorrect. This is the value of p that satisfies the equation $2p - 275 = 325$, not $2p + 275 = 325$.

Q8

C

C. 18

Choice C is correct. It's given that $4x = 3$. Multiplying each side of this equation by 6 yields $24x = 18$. Therefore, the value of $24x$ is 18.

Choice A is incorrect. This is the value of $6x$, not $24x$.

Choice B is incorrect. This is the value of $8x$, not $24x$.

Choice D is incorrect. This is the value of $96x$, not $24x$.

Q9 B

B. 130

Choice B is correct. It's given that the film club has 90 members on the first day of a semester, and 10 new members join the film club each day after the first day of the semester. This means that after 4 days, 4×10, or 40, new members will have joined the club. Adding 40 members to the original 90 club members yields 130 members. Thus, the film club will have 130 total members 4 days after the first day of the semester.

Choice A is incorrect. This is the number of members that will have joined the film club 4 days after the first day of the semester if 100 new members, not 10, join the film club each day.

Choice C is incorrect. This is the number of members the film club will have 4 days after the first day of the semester if 1 new member, not 10, joins the film club each day.

Choice D is incorrect. This is the number of members the film club has on the first day of the semester.

Q10 .2

Accept also: 1/5

The correct answer is $\frac{1}{5}$. Since the number 5 can also be written as $\frac{5}{1}$, the given equation can also be written as $\frac{x}{8} = \frac{5}{1}$. This equation is equivalent to $\frac{8}{x} = \frac{1}{5}$. Therefore, the value of $\frac{8}{x}$ is $\frac{1}{5}$. Note that 1/5 and .2 are examples of ways to enter a correct answer.

Alternate approach: Multiplying both sides of the equation $\frac{x}{8} = 5$ by 8 yields $x = 40$. Substituting 40 for x into the expression $\frac{8}{x}$ yields $\frac{8}{40}$, or $\frac{1}{5}$.

Notes

Accept equivalent forms (e.g. `0.25` ≡ 1/4). For grid-in items, decimal and fraction answers are both valid.

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